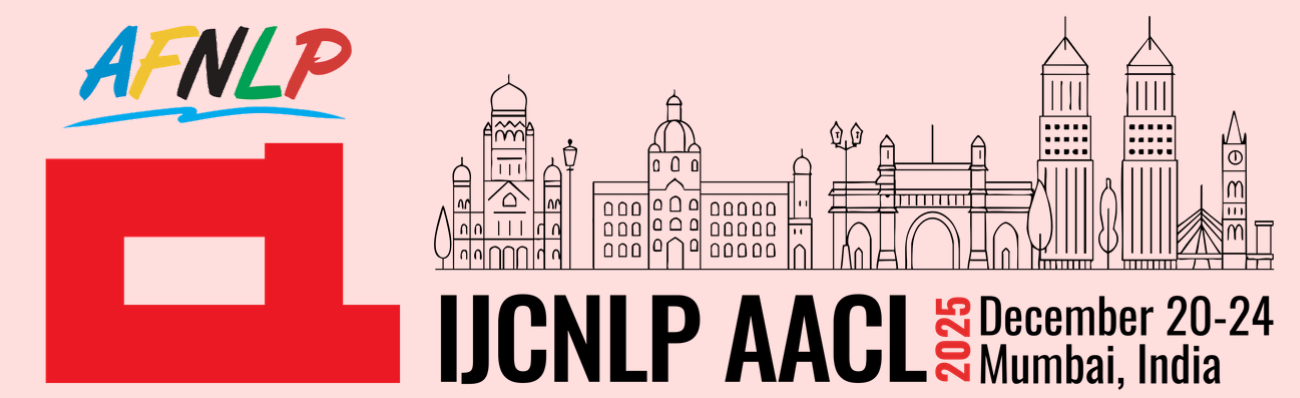


Iterative Critique-Driven Text Simplification: Targeted Enhancement of Complex Definitions with Small Language Models

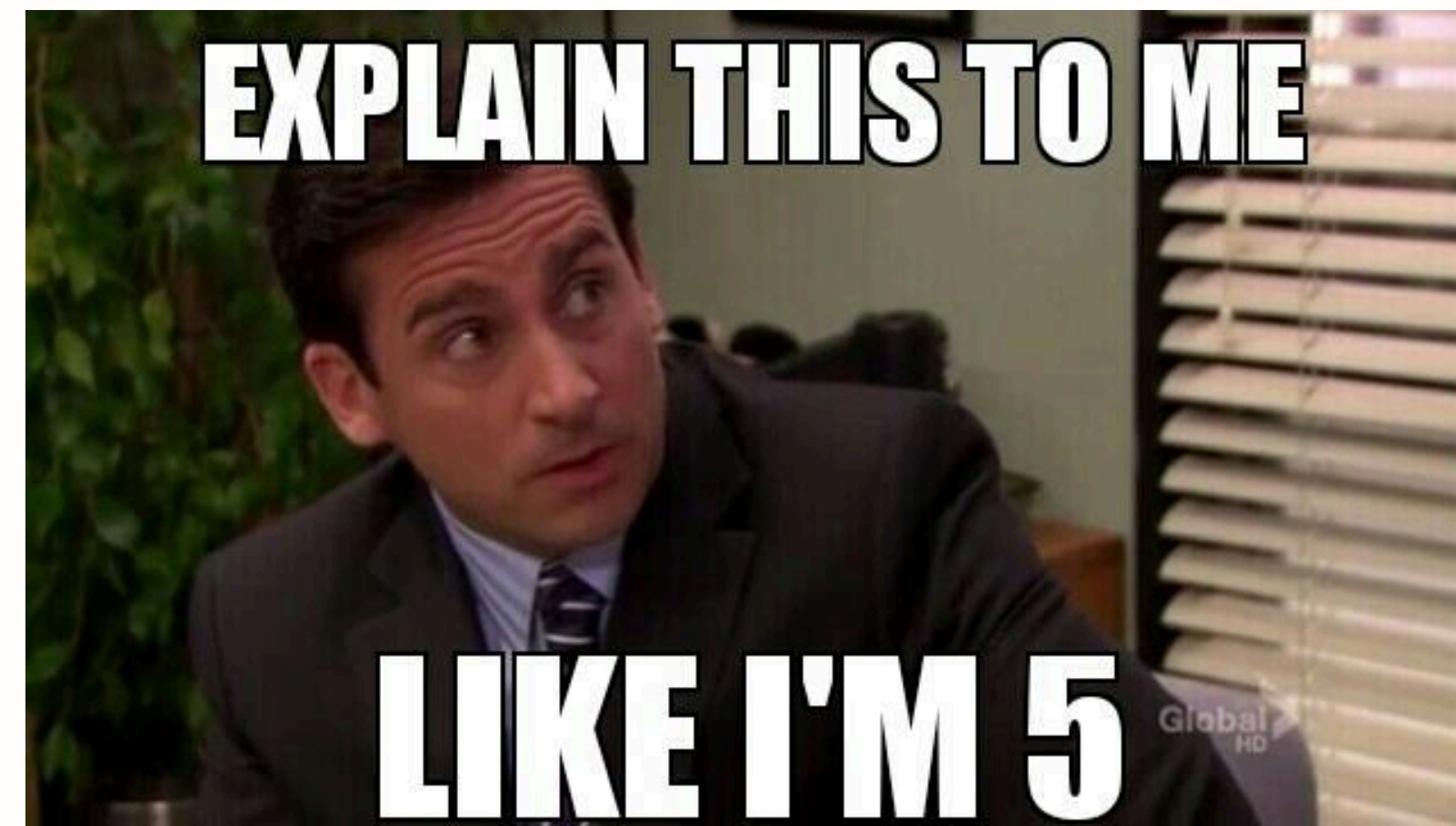
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The Problem

Difficult and unfamiliar concepts (like ‘*electron gun*’) often hinder comprehension for lay audiences, especially in technical and educational domains. This motivates the usage of language models for text simplification.



The Goal: Reduce cognitive load without losing technical precision.

The Solution

We propose:

- An **iterative refinement framework** that aims to simplify definitions by carefully handling complex terminology and domain-specific expressions.
- The obtained definition is **reprocessed based on the critique**, making refinements in successive iterations.

SLM Focus: SLMs allow for fast, modular agents where specific small models handle specific sub-tasks efficiently and in a cost-effective manner.

Implementation Details

Data: We used the **WIKIDOMAINS** Dataset with 22,561 domain-specific definitions spanning 13 academic domains from Wikipedia, with each definition having a *difficult concept*.

Models:

- **Small Language Models:** Gemma-3 (1B, 4B), LLaMA-3.2 (1B, 3B), Qwen-3 (1.7B, 4B)
- **GPT-4o:** SOTA baseline, critic and LLM-as-a-judge

Comparative baselines: Plain rewrite, Iteration 1 and Iteration 2

But what does the critic look for? We present the following rubrics focused on cognitive accessibility:

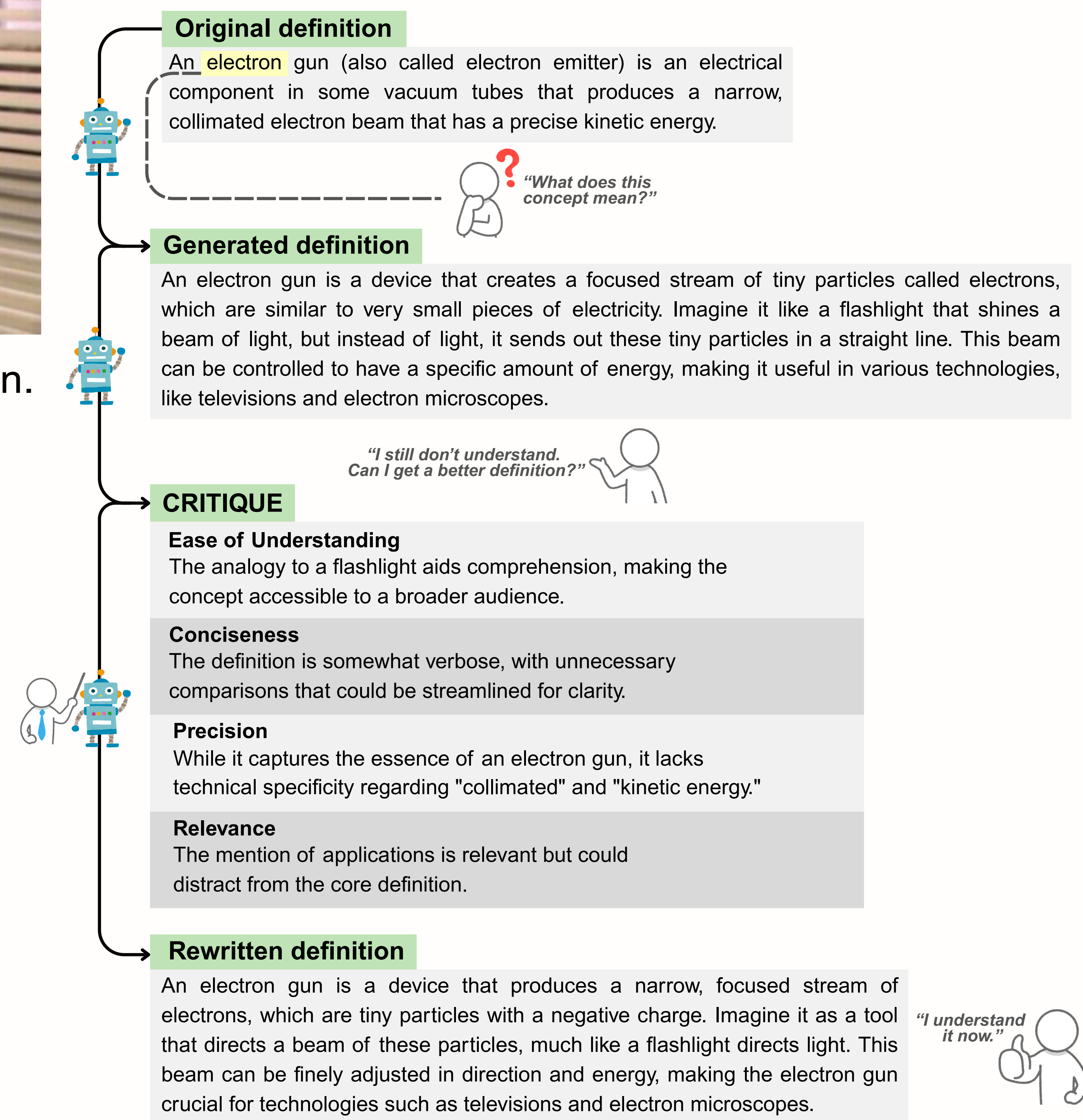
Criterion	Description
Ease of Understanding	Is it easy for someone with no background to follow?
Precision	Is the explanation specific and correct? Avoid vagueness or technical jargon.
Conciseness	Is it clear and brief, without unnecessary elaboration?
Relevance	Does it stick closely to the original definition without adding unrelated ideas?

- Instead of using a particular strategy (*simplify/explain/give an example/use an analogy*), our framework is **strategy agnostic**.
- The model uses a strategy appropriate to the definition, which is then refined by critic LLM. This allows for more flexible generations based on the context of the definition.

Results

- ‘*Base*’ indicates initial rewrite, ‘*I1*’ and ‘*I2*’ indicate the following iterations of the rewritten definition.
- Across Llama 3.2, Gemma 3, and Qwen 3, we observed **consistent gains in semantic fidelity**, with Llama 3.2 1B improving its BERTScore from 0.83 to 0.94 after just two iterations.
- Notably, Qwen 3 4B achieved ‘**Relevance**’ and ‘**Precision**’ scores comparable to **GPT-4o**, validating that high-quality simplification is achievable on consumer-grade hardware.
- Together, these findings suggest the critique loop is particularly effective at reigning in “hallucinations” or tangential wanderings, forcing the SLM to stay on topic and producing cognitively efficient simplified results.

Methodology



An example of a definition from **WIKIDOMAINS** dataset containing a potentially ‘**difficult concept**’, refined through iteration using critique-based LLM feedback.

Human Validation

We conducted a human survey using a total of **18 non-expert** volunteer annotators. We sampled a set of 50 definitions from the test set preserving the original domain distribution.

EL	I	EoU	Conciseness	Relevance	Precision
Bachelor’s	Base	3.25 ± 1.51	3.05 ± 1.31	3.22 ± 1.45	3.17 ± 1.29
	I1	3.13 ± 1.54	3.13 ± 1.32	3.14 ± 1.47	3.25 ± 1.31
	I2	3.35 ± 1.59	3.33 ± 1.30	3.43 ± 1.44	3.55 ± 1.29
High School	Base	4.35 ± 0.84	3.88 ± 0.98	4.24 ± 0.81	3.96 ± 0.83
	I1	4.34 ± 0.72	3.93 ± 0.91	4.28 ± 0.75	4.20 ± 0.77
	I2	4.36 ± 0.72	4.03 ± 0.87	4.34 ± 0.71	4.30 ± 0.68
Master’s	Base	4.33 ± 0.78	3.76 ± 1.19	4.07 ± 0.88	4.24 ± 0.75
	I1	4.26 ± 0.88	3.76 ± 1.14	4.00 ± 0.78	4.20 ± 0.84
	I2	4.31 ± 0.77	3.70 ± 1.04	4.01 ± 0.74	4.21 ± 0.76
PhD	Base	3.81 ± 1.20	3.50 ± 0.83	3.70 ± 0.75	3.60 ± 0.89
	I1	4.01 ± 1.10	3.62 ± 1.01	3.59 ± 0.87	3.68 ± 0.96
	I2	4.02 ± 1.07	3.62 ± 0.85	3.69 ± 0.84	3.70 ± 0.87

Mean ± standard deviation of survey responses across iterations and education levels for four text quality dimensions. (**EoU** - *Ease of Understanding*, **EL** - *Education Level*, **I** - *Iteration*)